

Mental Health and Social Media Balance Report

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Introduction

The rapid rise of smartphones and social media has raised important questions about how digital behavior influences mental health. As platforms become central to daily communication and entertainment, researchers increasingly examine how usage patterns relate to sleep, stress, happiness, and other psychological outcomes. Prior work links intensive digital media use to disrupted sleep, anxiety, decreased happiness and other psychological challenges (Korte, 2020), highlighting the importance to understand which aspects of social media use are most strongly associated with mental-health outcomes.

In this project, we analyze a Kaggle dataset (Rajak, 2023), of 500 users containing demographic information, lifestyle habits, social media behavior, and self-reported ratings of sleep quality, stress level, and happiness. We apply three machine learning approaches:

- Unsupervised clustering to identify lifestyle-based subgroups
- Supervised regression to predict mental-health scores from social media usage
- Supervised classification to predict platform type from mental-health and lifestyle metrics

Together, these methods provide a holistic picture of how mental health, lifestyle behaviors, and social media engagement interact.

Data and Exploratory Analysis

We began by familiarizing ourselves with the data by determining the dataset consists of 500 users and nine variables: Age, Gender, Daily Screen Time (hours), Sleep Quality (1–10), Stress Level (1–10), Days Without Social Media, Exercise Frequency (per week), Social Media Platform, and Happiness Index (1–10). There are no missing values, and all mental-health and lifestyle metrics are numeric. Table 1 provides the descriptive statistics for the continuous variables (see Table 1).

	Age	Daily_Screen_Time(hrs)	Sleep_Quality(1-10)	Stress_Level(1-10)	Days_Without_Social_Media	Exercise_Frequency(week)	Happiness_Index(1-10)
count	500.000000	500.000000	500.000000	500.000000	500.000000	500.000000	500.000000
mean	32.988000	5.530000	6.304000	6.618000	3.134000	2.448000	8.376000
std	9.960637	1.734877	1.529792	1.542996	1.858751	1.428067	1.524228
min	16.000000	1.000000	2.000000	2.000000	0.000000	0.000000	4.000000
25%	24.000000	4.300000	5.000000	6.000000	2.000000	1.000000	7.000000
50%	34.000000	5.600000	6.000000	7.000000	3.000000	2.000000	9.000000
75%	41.000000	6.700000	7.000000	8.000000	5.000000	3.000000	10.000000
max	49.000000	10.800000	10.000000	10.000000	9.000000	7.000000	10.000000

Table 1: Descriptive statistics of variables

To better understand the structure of the dataset and identify potential patterns, we first visualized the distributions of several key features.

The bar chart of social media platform mentions (Figure 1) shows that platforms such as TikTok, X/Twitter, LinkedIn, Facebook, YouTube, and Instagram are all well represented, confirming adequate class balance for later modeling. The histogram of stress levels (Figure 2) displays a roughly normal distribution centered around scores of 6–8, indicating that moderate to high stress is common across users. Likewise, the distribution of daily screen time (Figure 3) appears approximately normal, with most individuals reporting between 4 and 7 hours of screen use per day.

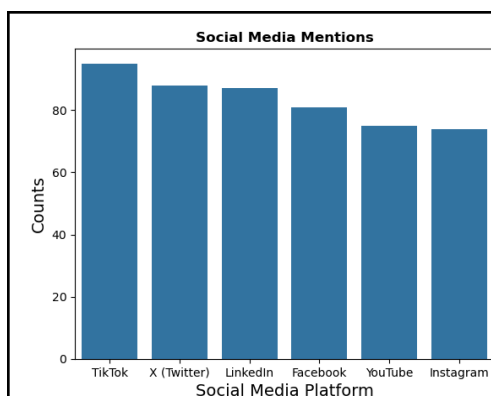


Figure 1: Social media platform counts.

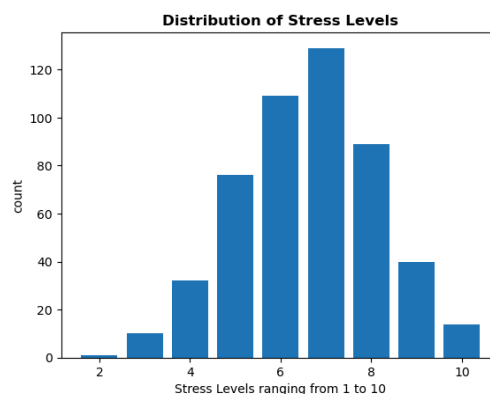


Figure 2: Stress level distribution

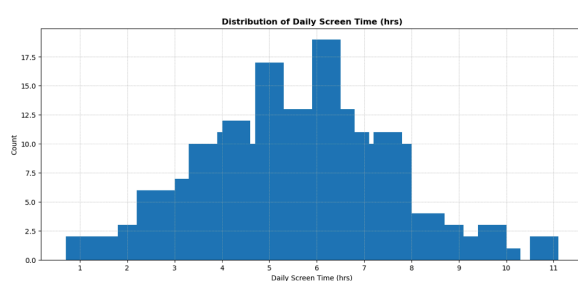


Figure 3: Daily screen time distribution

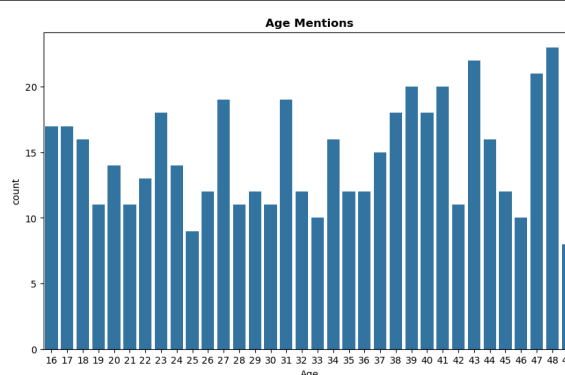


Figure 4: Stress by age group

Unsupervised Learning: Cluster Analysis (K-Means and GMM)

Research Question: What patterns emerge when we cluster users based on lifestyle and mental-health attributes, and how do these patterns relate to stress and happiness?

Theoretical and Practical Explanation

Clustering represents an unsupervised learning problem: we do not specify outcomes, but instead seek natural groupings within the data. We use K-Means to partition users into clusters with similar mean values across selected attributes, and Gaussian Mixture Models (GMM) as a probabilistic alternative that allows for overlapping clusters.

Practically, we are interested in whether combinations of daily screen time, sleep, stress, and exercise align into recognizable lifestyle profiles. If certain clusters combine high happiness, low stress, and healthier usage patterns, this could inform digital-well-being recommendations and highlight behaviors associated with better mental health.

Data and Preprocessing

For the clustering models the attributes: Daily Screen Time, Sleep Quality, Stress Level, Days Without Social Media, Exercise Frequency, and Happiness Index were utilized. All are continuous or quasi-continuous. Before fitting both K-Means and GMM, we standardized these features using the StandardScaler tool because these features are on different scales.

Model Performance

We fit K-Means clustering models using $k = 3$ to 8 clusters and evaluated model quality using inertia (within-cluster sum of squares) and silhouette scores. The Elbow Method plot (Figure 5) shows a noticeable bend at $k = 5$, where reductions in inertia begin to taper, indicating diminishing returns from additional clusters. The silhouette score (Figure 6) also peaks around $k = 5$, albeit at modest levels, suggesting that while clusters are distinguishable, some overlap between groups is expected. Taken together, these metrics support selecting five clusters as the optimal solution.

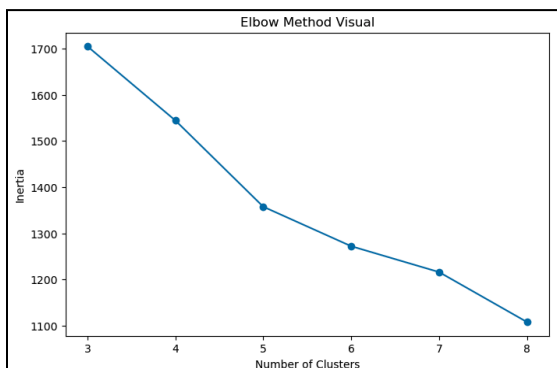


Figure 5: Elbow plot showing inertia values for K-Means clustering across $k = 3$ –8.

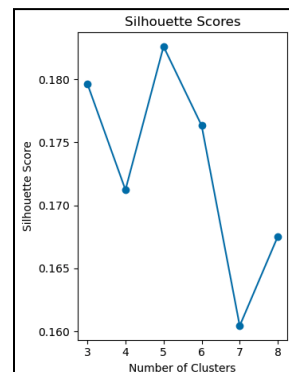


Figure 6: Silhouette scores for K-Means clustering across $k = 3$ –8.

To compare against a probabilistic approach, we fit Gaussian Mixture Models (GMMs) across the same range of cluster counts. We evaluated model fit using both the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). As shown in Table 2, both AIC and BIC reach their lowest values at $k = 5$, aligning with the K-Means results.

k	AIC	BIC
3	5281.739728	5631.552200
4	5282.337623	5750.159122
5	4260.730858	4846.561384
6	4275.729568	4979.569120
7	4316.151714	5138.000293

Table 2: AIC and BIC values for GMM models ($k = 3$ –7).

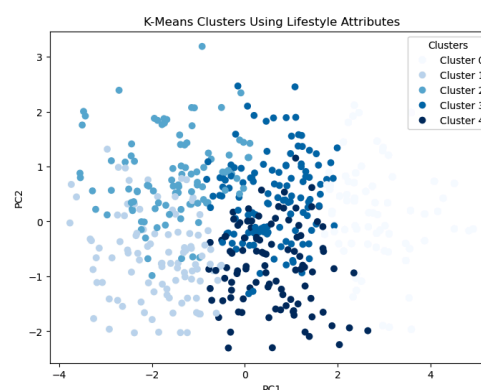


Figure 7: Scatterplot of the five K-Means clusters projected into two principal components.

To visualize the resulting clusters, we projected the standardized features into two principal components and plotted the cluster assignments. The scatterplot in Figure 7 displays five clusters with visible structure, though with expected overlap based on the silhouette values. This visualization confirms that lifestyle and mental-health variables form meaningful patterns in the dataset.

Cluster Profile and Findings

The mean feature values for each K-Means cluster are presented in Table 3. Two clusters (Clusters 1 and 2) stand out for their high happiness index, low stress levels, better sleep quality, and lower daily screen time, indicating generally healthier lifestyle patterns. In contrast, Cluster 0 exhibits the lowest happiness, highest stress, poorest sleep, and heaviest screen time (~8 hours per day), reflecting a high-risk profile for negative mental-health outcomes. Clusters 3 and 4 fall between these extremes, with moderate levels across most variables.

Cluster Label	Daily Screen Time (hrs.)	Sleep Quality (1-10)	Stress Level (1-10)	Days Without Social Media	Exercise Frequency	Happiness Index (1-10)
0	7.96	4.34	8.66	2.77	2.43	5.99
1	3.92	7.78	5.07	3.17	1.37	9.74
2	4.09	7.33	5.61	3.36	4.09	9.64
3	5.99	5.88	7.05	1.56	2.30	7.93
4	6.19	5.81	7.11	5.05	2.14	8.09

Table 3: K-Means cluster means for six lifestyle and mental-health attributes.

The GMM cluster means (Table 4) reveal highly similar patterns. Again, clusters with the highest happiness and lowest stress also show better sleep quality and reduced screen time, while the cluster with the lowest happiness shows the highest stress, lowest sleep, and heaviest screen use.

Across both K-Means and GMM, we observe a consistent lifestyle gradient:

- **Healthier clusters** → higher sleep quality, higher happiness, lower stress, and lower screen time.
- **Less healthy clusters** → lower sleep quality, lower happiness, higher stress, and substantially higher screen time.

These unsupervised findings provide an important foundation for the supervised regression and classification models, which examine these relationships in a more targeted way.

Cluster Label	Daily Screen Time (hrs.)	Sleep Quality (1-10)	Stress Level (1-10)	Days Without Social Media	Exercise Frequency	Happiness Index (1-10)
0	7.81	4.12	8.77	3.48	2.09	5.70
1	5.26	6.36	6.40	3.20	2.28	9.00
2	4.08	7.60	5.18	3.30	2.62	10.00
3	6.27	5.47	7.19	1.52	2.29	7.32
4	6.20	6.18	7.48	4.38	2.73	7.62

Table 4: GMM cluster means for six lifestyle and mental-health attributes.

Supervised Learning: Predicting Mental Health Outcomes (Linear Regression and Random Forest)

Research Question: Can social media usage patterns predict variations in mental-health outcomes (sleep quality, stress level, and happiness), and which indicators are the strongest predictors?

Theoretical and Practical Explanation

This research question represents a supervised regression problem. The three mental-health outcomes - sleep quality, stress level, and happiness - are measured on 1–10 scales that, while ordinal, behave closely enough to continuous variables to justify regression modeling. To capture different types of relationships within the data, we apply two complementary approaches: Linear Regression/OLS, which provides interpretable coefficients and significance testing, and Random Forest Regression, which captures non-linear patterns and interactions that linear models may miss. Together, these methods allow us to identify which social media behaviors are most predictive of mental-health variation and whether those relationships are primarily linear or more complex.

Practically, understanding how digital behavior relates to mental health is increasingly important as smartphones and social media have become woven into daily life. Widespread smartphone adoption began less than two decades ago (Coulstring, 2023), and social media usage accelerated rapidly in the early 2000s (Ortiz-Ospina, 2019). In that short time, intensive digital media use has been linked to disrupted sleep, elevated stress, and reduced well-being (Korte, 2020). By analyzing this dataset, we can examine how specific behaviors, particularly screen time, relate to sleep, stress, and happiness. These insights support public-health communication, inform platform design decisions, and help users understand how everyday digital habits may influence mental-health outcomes.

Method 1

After preprocessing the data, we used a Pipeline to standardize features, encode the platform variable, and fit the regression models. To identify the strongest predictors, we calculated permutation importance scores, which highlighted daily screen time as the primary contributor. We then used OLS regression to confirm the significance of this predictor and compared performance metrics (R^2 and MAE) to evaluate how accurately the models could estimate each mental-health outcome using the most influential features.

Model	Features	Performance Metric	Notes
Linear Regression (Pipeline)	Daily Screen Time Days Without Social Media Social Media Platform	Sleep Quality: MAE = 0.73, R^2 = 0.56 Stress Level: MAE = 0.86, R^2 = 0.55 Happiness Index: MAE = 0.85, R^2 = 0.54	Good predictive accuracy, models explain about half of the variance of all outcomes. Predicated values closely match actual ratings.
OLS Regression (Significance Testing)	Same features as pipeline	Coefficients only significant for Daily Screen Time, $p < .001$	Each extra hour of screen time lowers sleep quality and happiness by ~1 point, and raises stress by ~1 point.
Permutation Importance (Feature Strength)	Same features as pipeline	Daily Screen Time mean drop in R^2 = 1.0, all others < 0.01	Confirms that screen time is the dominant predictor; other features contribute negligibly.

Correlation Analysis (Exploratory)	All variables	Daily Screen Time correlated with sleep (-0.76), stress (+0.74), and happiness (-0.71)	Relationships align with regression results: more screen time = worse sleep and happiness, higher stress
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Method 1 Results

Both the scikit-learn Pipeline and the OLS models consistently demonstrated that Daily Screen Time is the strongest and most significant predictor of self-reported mental health outcomes. However, the models achieved a moderate R^2 of approximately 0.55 across all three targets (Sleep Quality, Stress Level, and Happiness Index). This finding is crucial: while the models account for a significant portion of the variance (about 55%), it also means that nearly 45% of the variation in these mental health scores remains unexplained by the measured social media features (daily time, platform, and days away). This high unexplained variance strongly suggests that external, unmeasured factors play a substantial role in determining an individual's mental well-being.

Method 2

We ran a Random Forest Regression model on the data. This data had three outcomes: Sleep Quality, Stress Level, and Happiness Index that were all indicative of mental health outcomes. To determine which were the strongest predictors we found the importance scores of the attributes and then ran additional models using only the strongest predictors and pulled metrics [R^2 , MAE] to compare the Random Forest Model's prediction accuracy based on the specified predictors.

Model	Features	Performance Metric	Notes
Random Forest [3 Models]	Trial #1: Daily Screen Time, Days Without Social Media, Social Media Platform Trial #2: Daily Screen Time, Age, Gender, Weekly Exercise Frequency, Days Without Social Media, Social Media Platform Trial #3: Daily Screen Time Outcome Variables: Sleep Quality, Stress Level, Happiness Index	We pulled the R^2 and MAE values for each trial for the three outcome variables: Stress Level, Sleep Quality, and Happiness Index. Please refer to the table in the results for further information.	The model using only Daily Screen Time had the highest R^2 values out of any of the trials for all three outcome variables. The MAE values for the same model was the lowest for all three outcome variables in trial #3. We ran a Random Forest Regression model as we had continuous data instead of categorical data.

Method 2 Results

The first model that was run included the daily screen time in hours, days without social media and specific social media platform to determine the outcome variables [stress level, sleep quality, and happiness index]. The importance scores (IS) were determined for these models as shown below:

Outcome Variable	Daily Screen Time (Hrs) IS	Days without social media IS	Social Media platform – Twitter IS	Social Media platform – LinkedIn IS	Social Media platform – TikTok IS	Social Media platform – Instagram IS	Social Media platform – Youtube IS
Sleep Quality	0.773	0.117	0.025	0.024	0.022	0.021	0.018
Stress Level	0.754	0.132	0.027	0.024	0.023	0.022	0.018

Happiness Index	0.734	0.141	0.028	0.026	0.026	0.025	0.021

Table 5: Random Forest Regression Trial #1 Importance Scores

The table above showed that for every mental health outcome variable the daily screen time was the most important variable when predicting the outcomes. This value was always at least 5.2x more ‘important’ than the highest importance score after it.

To determine if any of the variables in the mental health data set showed a high importance value a second Random Forest model was run for all the predictors: Daily Screen Time, Age, Days without Social Media, Weekly Exercise Frequency, Gender, and Social Media Platform [TikTok, Twitter, LinkedIn, Instagram, Youtube]. The importance scores for this model are as listed below per mental health outcome variable:

Outcome Variable	Daily Screen Time (Hrs) IS	Age IS	Days without Social Media IS	Weekly Exercise Frequency IS	Gender [Male] IS	Social Media platform – TikTok IS	Social Media platform – Twitter IS	Social Media platform – LinkedIn IS	Social Media platform – Instagram IS	Social Media platform – Youtube IS	Gender [Other] IS
Sleep Quality	0.687	0.113	0.060	0.056	0.016	0.014	0.013	0.011	0.011	0.010	0.008
Stress Level	0.660	0.109	0.074	0.061	0.017	0.016	0.016	0.015	0.015	0.011	0.006
Happiness Index	0.641	0.128	0.0174	0.055	0.018	0.018	0.018	0.016	0.014	0.012	0.007

Table 6: Random Forest Regression Model Trial #2 Importance Scores

This table shows that the Daily Screen Time was the most important predictor to determine any of the outcome variables. The Daily Screen Time predictor had an importance score 5x greater than the next highest importance score for any of the outcome trials.

The R^2 and MAE values were run for each of these trials and one additional trial to determine how the models were performing:

Trial # : Outcome Variable	R^2	MAE
<i>1: Sleep Quality</i>	0.441	0.841
<i>1: Stress Level</i>	0.493	0.889
<i>1: Happiness Index</i>	0.427	0.897
<i>2: Sleep Quality</i>	0.496	0.764
<i>2: Stress Level</i>	0.498	0.898
<i>2: Happiness Index</i>	0.454	0.904
<i>3: Sleep Quality</i>	0.498	0.768
<i>3: Stress Level</i>	0.540	0.850
<i>3: Happiness Index</i>	0.501	0.847

Trial 1 Predictors: Daily Screen Time (hrs), Days without Social Media, Social Media Platform

Trial 2 Predictors: Daily Screen Time (hrs), Days without Social Media, Age, Weekly Exercise Frequency, Gender, and Social Media Platform

Trial 3 Predictors: Daily Screen Time (hrs)

The R^2 was the highest for the Random Forest model trial 3 meaning that 49.8%, 54.0%, and 50.1% of the variance in the respective outcome variables is explained by the predictor: Daily Screen Time. Overall the Daily Screen Time related to the mental health outcomes of the users in the data set, but there are additional variables related to their mental health outcomes that were not measured in the

dataset. We know that this is the case because when we used all the predictor variables in the data set for the mental health outcomes our R^2 values were never higher than 49.3% of any of the outcome variables.

The dataset showed that there are other attributes that are contributing to people's measurable mental health outcomes and social media usage is playing a part, but these other factors are contributing at the minimum 46% of the variance we see in stress levels, 49.2% of the variance we see in the happiness index, and 50.1% of the variance we see in sleep quality for the users in the data set.

Method Comparison

Across both modeling approaches, the results consistently showed that daily screen time was the strongest and most important predictor of all three mental-health outcomes. In the linear regression analyses, daily screen time was the only statistically significant variable in the OLS models, with $p < .001$ for sleep, stress, and happiness. The pipeline models achieved R^2 values around 0.54–0.56, meaning the predictors explained just over half of the variation in people's 1–10 mental-health scores. Permutation importance further confirmed these findings: removing daily screen time caused the model's accuracy to drop sharply, while removing any other predictor had almost no impact.

The Random Forest models reinforced this pattern. Feature-importance tables show that daily screen time consistently had the largest importance score, often five times larger than any other variable. When we compared model performance across Random Forest trials, the model that used only daily screen time had the highest R^2 values and the lowest MAE for all three outcomes. Even after adding additional predictors such as age, weekly exercise frequency, and gender, the overall predictive power did not improve.

Together, these results show strong agreement between the two methods. Both Linear Regression and Random Forest Regression highlight daily screen time as the primary driver of differences in sleep quality, stress level, and happiness ratings. At the same time, both approaches show that only about half of the variation in these mental-health outcomes can be explained by the variables in the dataset. This suggests that while social media usage (especially total screen time) plays an important role, many other influences on mental health were not captured in the data.

Supervised Learning: Platform Classification (Logistic Regression)

Research question: Can we classify a user as belonging to a video-based or text-based social media platform group based on their mental health and lifestyle metrics?

Theoretical and Practical Explanation

This task is a supervised binary classification problem. Our target variable indicates whether a user's primary platform is video based or text based. We used logistic regression, which models the log-odds of class membership as a linear function of predictors. Logistic regression is appropriate because it provides interpretable coefficients and probabilistic outputs, enabling us to see not only whether mental-health variables predict platform type, but also how each feature shifts the probability of belonging to the video group. Its transparency complements the more flexible but less interpretable models used in earlier sections.

Practically, understanding how well-being patterns relate to platform type can inform design decisions and interventions. Video-centric platforms often encourage immersive, continuous scrolling, whereas text-oriented platforms typically support shorter, more fragmented interactions. If mental-health

profiles differ between these groups, that has implications for how we think about digital-well-being tools and platform responsibility.

Data, Preprocessing, and Modeling

For this model we used six predictors: Daily Screen Time, Sleep Quality, Stress Level, Days Without Social Media, Exercise Frequency, and Happiness Index. These variables capture the same lifestyle and mental-health dimensions used in our other analyses.

We recorded Social Media Platform into a binary label (video vs. text). Platforms that did not clearly fit either category were excluded. Using scikit-learn, we built a pipeline that standardized all predictors and fit a LogisticRegression classifier with `max_iter = 1000` and `class_weight = "balanced"` to address mild class imbalance.

Model Performance and Interpretation

METRIC	Accuracy	Precision	Recall	F1-score	ROC-AUC
VALUE	0.570	0.568	0.510	0.538	0.597

Table 7: Logistic Regression Metrics

On the test set, the logistic regression model achieved the metrics shown in Table 7: accuracy = 0.57, precision = 0.57, recall = 0.51, F1-score = 0.54, and ROC-AUC = 0.60. These values indicate moderate predictive ability, above random guessing but far from perfect. The confusion matrix (Figure 8) shows that the model classifies text and video users at similar rates (32 text and 25 video correctly labeled), suggesting no strong bias toward either class. The ROC curve (Figure 9) lies consistently above the diagonal, confirming that the model ranks video users above text users more often than not.

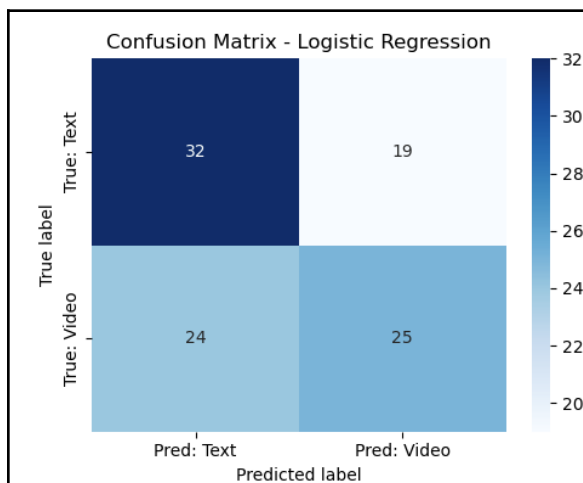


Figure 8: Confusion Matrix.

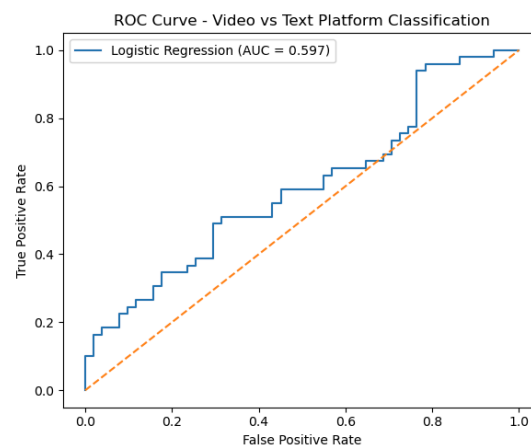


Figure 9: ROC Curve

Standardized coefficients (Table 8) reveal how each feature influences the probability of using a video-based platform. Exercise Frequency is the strongest positive predictor, indicating that more active users are somewhat more likely to favor video platforms. Sleep Quality and Daily Screen Time also increase the likelihood of video use, though more modestly. Happiness Index is the strongest negative predictor, suggesting that less happy users gravitate toward video platforms, while happier users are more likely to prefer text-based platforms. Stress Level and Days Without Social Media have minimal

coefficients, indicating little role in distinguishing platform type.

Feature	Coefficient	Interpretation
Exercise_Frequency(week)	+0.232	Higher exercise frequency is associated with a greater likelihood of video-platform use.
Sleep_Quality(1–10)	+0.121	Better sleep quality slightly increases the probability of video-platform preference.
Daily_Screen_Time(hrs)	+0.071	Users with more daily screen time are more likely to use video-based platforms.
Days_Without_Social_Media	+0.017	Minimal impact on platform type.
Stress_Level(1–10)	+0.014	Minimal impact: stress does not strongly differentiate platform preference.
Happiness_Index(1–10)	–0.219	Higher happiness levels are associated with a greater likelihood of text-platform use.

Table 8: Standardized coefficients

Overall, mental-health and lifestyle metrics contain a detectable but limited signal for predicting whether someone’s primary platform is video or text based. The modest performance is expected given that platform choice is influenced by many other factors, yet the patterns in happiness, exercise, and screen time align with our earlier findings.

Conclusion

Using a combination of unsupervised clustering, linear and non-linear regression, and logistic regression, we examined how social media usage patterns relate to mental health and platform preference in a sample of 500 users. Across all methods, daily screen time consistently emerged as the dominant predictor of mental-health outcomes, with higher usage linked to poorer sleep, elevated stress, and lower happiness. The clustering analysis further revealed that groups characterized by healthier profiles; better sleep, higher happiness, and lower stress, also tended to have substantially lower screen time. Platform type proved more difficult to classify. Although logistic regression performed moderately above chance, it indicated that users of video-based platforms generally reported higher screen time, lower happiness, and higher exercise frequency compared to users of text-based platforms.

These results suggest that, while social media is only one piece of the mental-health puzzle, how much time people spend on screens, and on which platforms, matters. As digital technologies continue to evolve, data-driven analyses like this can help guide more thoughtful design, personal habits, and policy decisions that support mental well-being.

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